

Nama: Calvin Octo Fernando
NIM: 2802401335

Project: Honkai Star Retail

Frontend: Android (Flutter) using Flutter SDK version 3.32.2 and Android SDK API 35.

Backend: Node.js (version 22.16.0) + Express.

Database: MySQL via XAMPP.

User Roles:

- Admin: Has permission to add, edit, and delete resources.
- User: Has permission to view, search, filter, and purchase resources.

Database:

Table: users

- id (INT, PK, AutoIncrement)
- username (VARCHAR 100)
- email (VARCHAR 150)
- password (VARCHAR 255, bcrypt encrypted)
- role (ENUM 'admin' / 'user')
- google_id (VARCHAR 255)
- created_at (TIMESTAMP).

Table: resources

- id (INT, PK, AutoIncrement)
- name (VARCHAR 150)
- type (VARCHAR 100)
- description (TEXT)
- stock (INT)
- image (VARCHAR 255)
- price (DECIMAL 10,2)
- created_at (TIMESTAMP).

Table: purchases

- id (INT, PK, AutoIncrement)
- user_id (INT, FK)
- resource_id (INT, FK)
- quantity (INT)
- total_price (DECIMAL 10,2)
- purchased_at (TIMESTAMP).

Application Features:

Authentication:

- Supports login via email/password stored in the database, and Google OAuth (External OAuth).
- New account registration with validation.
- Generates a JWT bearer token (>20 alphanumeric characters).
- Verifies the bearer token on every request requiring authentication.
- Features a splash screen that automatically checks the login status.

Admin Features:

- Can view a list of all resources (GET).
- Can add new resources with images (POST).
- Can update existing resource data (PUT).
- Can delete resources, includes a confirmation dialog (DELETE).

User Features:

- Can view a list of resources in a grid layout (GET).
- Can search for resources by name and filter them by type (Light Cone, Material, Relic, Other).
- Can view full resource details and images.
- Can purchase resources by selecting the quantity via a bottom sheet (POST).
- Can view their purchase history (GET).

Data Validations:

- Emails must use a valid format containing an "@" symbol.
- Passwords must be at least 6 characters long and contain both letters and numbers.
- Password confirmation must match the entered password.
- Usernames must be at least 3 characters long.
- Stock must be a positive number, and the price must be greater than 0.
- Purchase quantities cannot exceed the available stock.
- The app displays error messages if validation fails.

Method	Endpoint	Description	Auth Required
POST	/api/auth/register	Register new user	No
POST	/api/auth/login	User login, generate JWT	No

POST	/api/auth/google	Google OAuth login	No
GET	/api/resources	Fetch all resources	Bearer Token
GET	/api/resources/:id	Fetch resource by ID	Bearer Token
POST	/api/resources	Add new resource (Admin)	Bearer Token
PUT	/api/resources/:id	Update resource (Admin)	Bearer Token
DELETE	/api/resources/:id	Delete resource (Admin)	Bearer Token
POST	/api/purchases/buy	Buy resource	Bearer Token
GET	/api/purchases/history	Fetch purchase history	Bearer Token

UI Components:

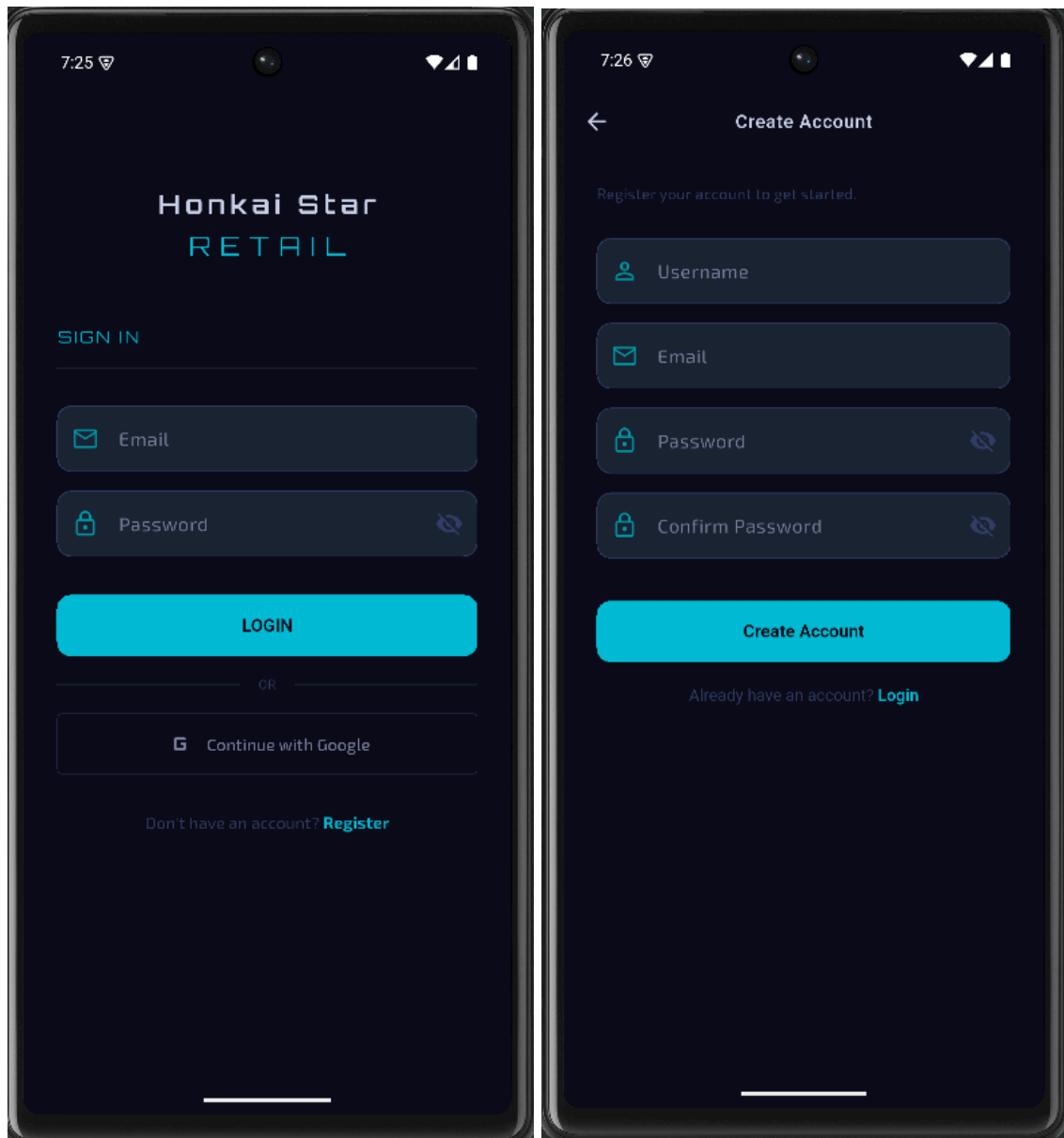
TextFormField, ElevatedButton, OutlinedButton, GridView (SliverGrid), ListView, DropdownButtonFormField, BottomSheet, CircleAvatar, AnimatedContainer, and AlertDialog.

Theme:

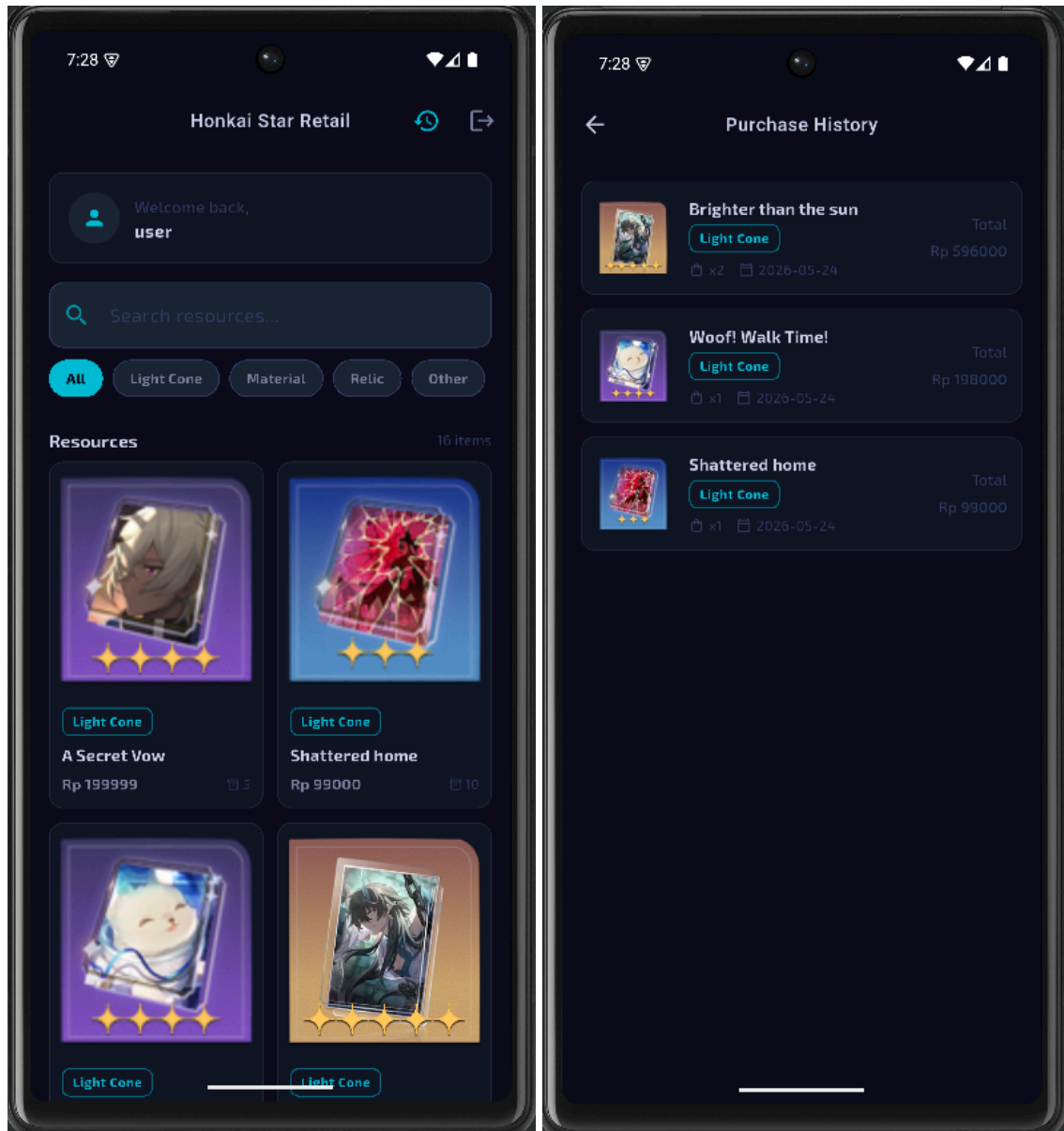
- Colors: Deep space-blue background (#0A0E1A), midnight surface containers (#111827, #1C2333), and vibrant cyan accents (#00BCD4) for active states, inputs, and buttons.
- Fonts: Global implementation of the Exo 2 font family with bold headers (FontWeight.w600) to maximize readability.
- Shapes: A uniform curve radius (BorderRadius.circular(12)) applied consistently across all form fields, resource cards, and buttons.

Application Screens:

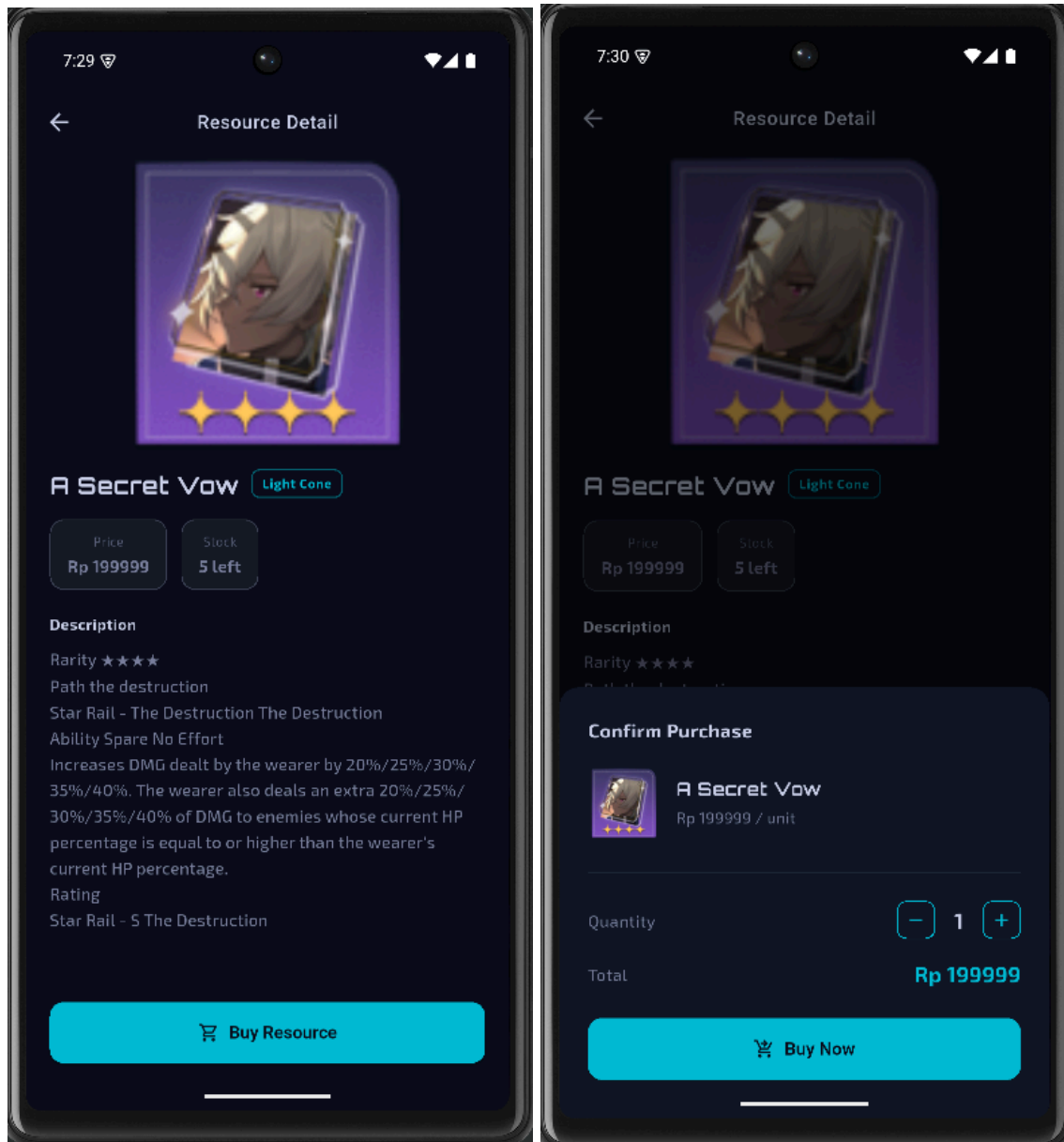
- Login Screen and Register Screen



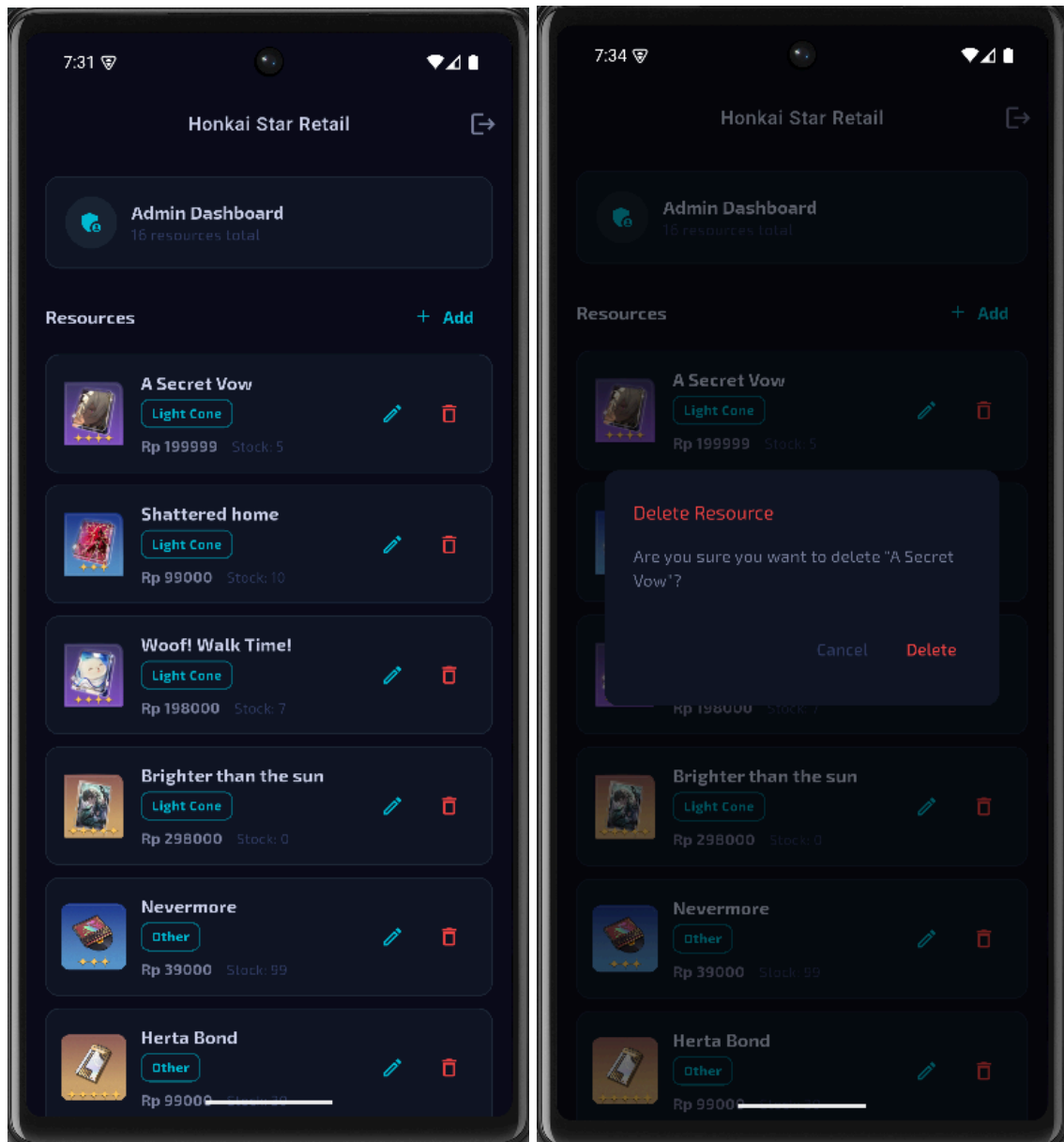
- Home Screen and Purchase History Screen



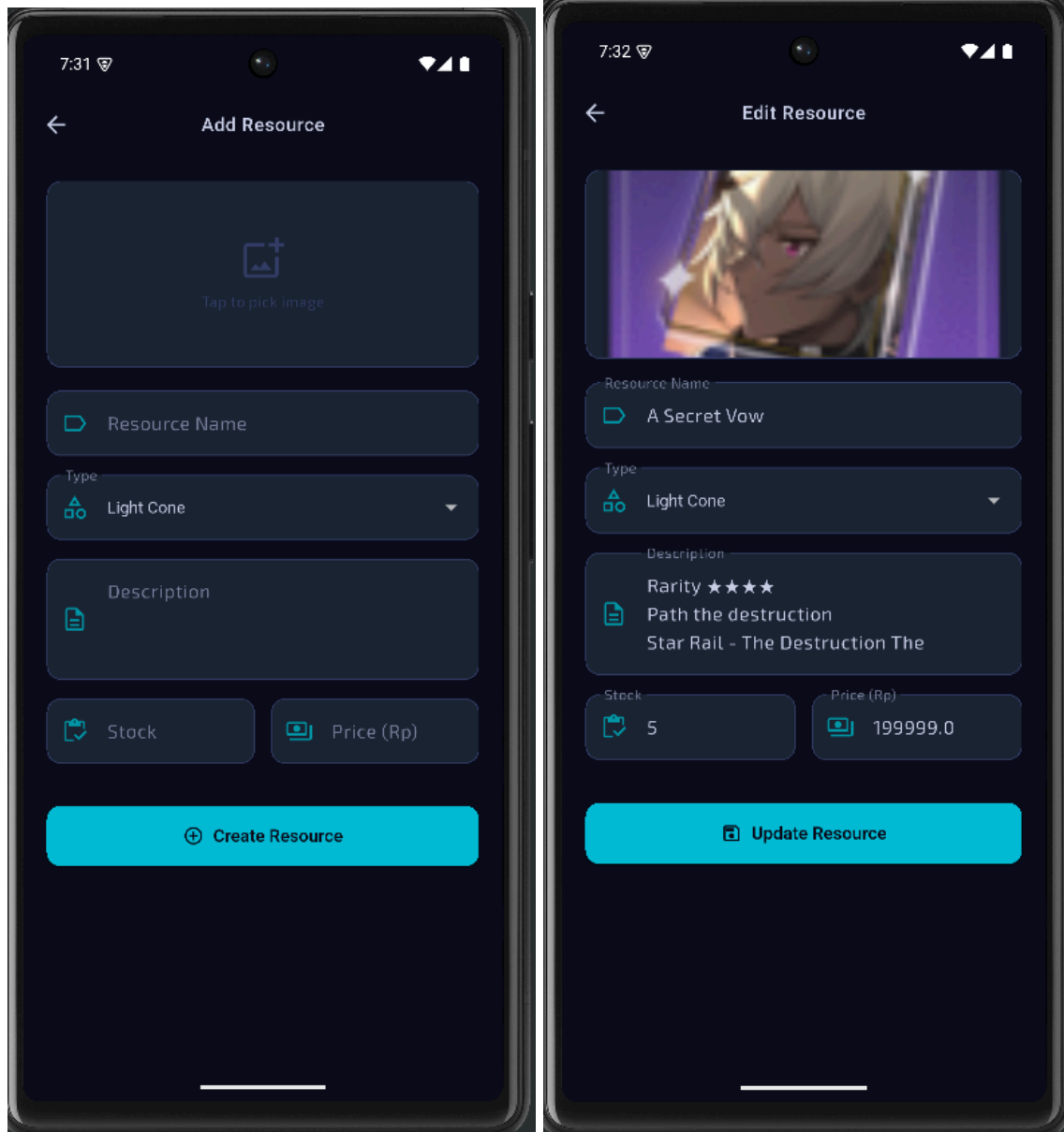
- Resource Detail Screen



- Admin Dashboard Screen and Delete Resource



- Admin Resource Form Screen And Update Resource



User:

1. Login (user@gmail.com || user123 or google)
2. Register if no account
3. After login, taken into home screen showing list of resources (can search or filter by category)
4. Tap resource to view resources full detail and tap buy resources
5. Select quantity and tap buy now to confirm purchase
6. In home screen, tap history icon to view the purchases history
7. Tap the logout icon to sign out.

Admin:

1. Login using the admin account (admin@gmail.com || admin123).
2. Taken to the Admin Dashboard.
3. Tap Add to add a new resource.
4. Fill in the form: name, type, description, stock, price, and image.
5. Tap Create Resource to save.
6. Tap the edit icon (pencil) on a resource to edit its data.
7. Tap the delete icon (trash) to delete a resource and confirm on the dialog.

Asset References

- Fonts: Orbitron and Exo 2 (<https://fonts.google.com/>)
- Images: Honkai: Star Rail game content owned by HoYoverse
- Libraries & Frameworks: Flutter SDK 3.32.2, Node.js 22.16.0, Express.js, Google Sign-In, JWT (jsonwebtoken), Multer, and bcryptjs.

Run apps:

- Database: import sql
- Backend: npm install – npm start / node [index.js](#)
- Frontend: flutter pub get – open emulator (api 35) – flutter run